

Resource Description Framework (RDF)

IE650 Knowledge Graphs

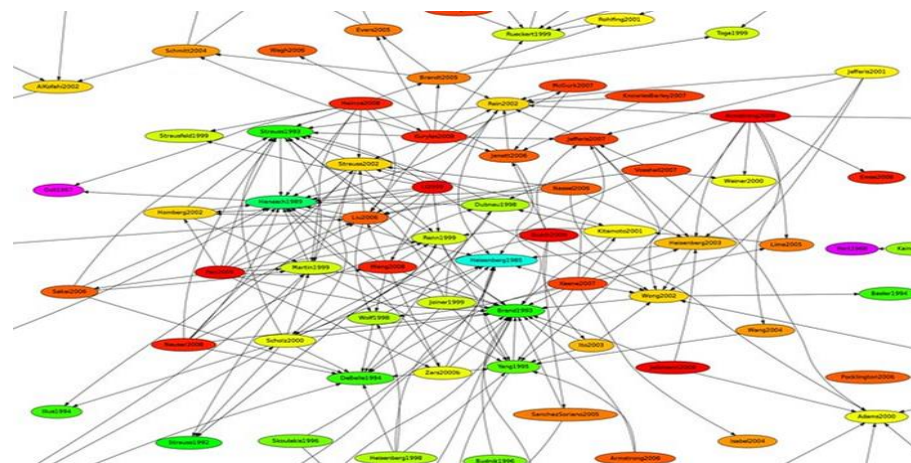


Overview

- Graph representation in a nutshell
- A brief history of RDF
- Encodings of RDF
- Semantics and principles of RDF
- Embedding RDF in HTML – RDFa, Microdata, Microformats
- RDF Tools

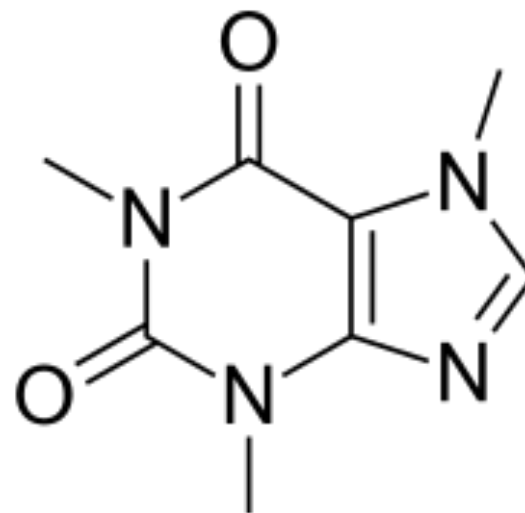
Knowledge Graphs are Graphs

- Graphs may have different flavours
- Directed vs. undirected, e.g.:
 - Social graph: A and B know each other
 - Citation graph: paper X cites paper Y



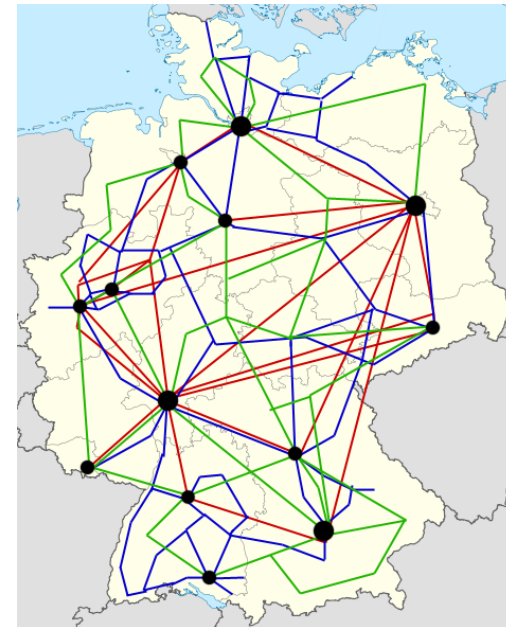
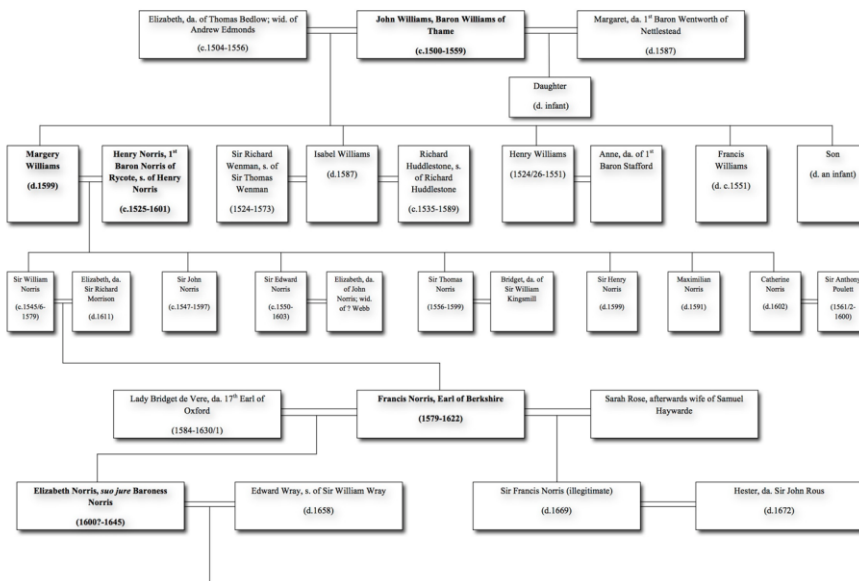
Knowledge Graphs are Graphs

- Graphs may have different flavours
- Labeled vs. unlabeled (also: heterogeneous vs. homogeneous)
 - A network of highways: a particular highway links to cities
 - A molecular graph



Knowledge Graphs are Graphs

- Graphs may have different flavours
- Cyclic vs. acyclic
 - A family tree (acyclic graphs are often referred to as “trees”)
 - A computer network



Knowledge Graphs are Graphs

- So when we talk about a knowledge graph, what kind of graph do we mean?
- No formal definition, but de facto consensus:
 - Directed, labeled graph
 - Heterogeneous node types (and edges)
 - Need not be cycle free
- Node types (“classes”) and edge types (“properties”)
 - Are also referred to the “schema” of the graph (aka “ontology”)
 - Can be defined with further restrictions
 - e.g., an edge of type “author” links a publication to a person

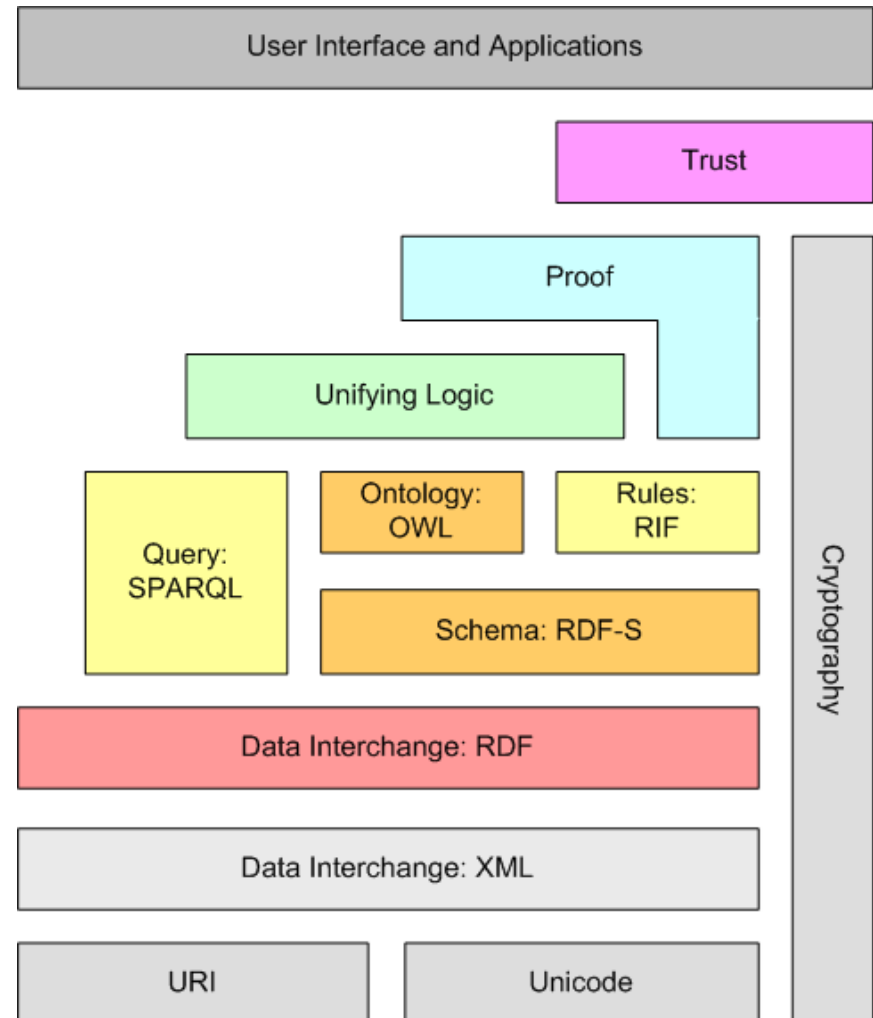
Semantic Web – Architecture



here be dragons...

Knowledge Graph Technologies
(This lecture)

Technical
Foundations



What is RDF?



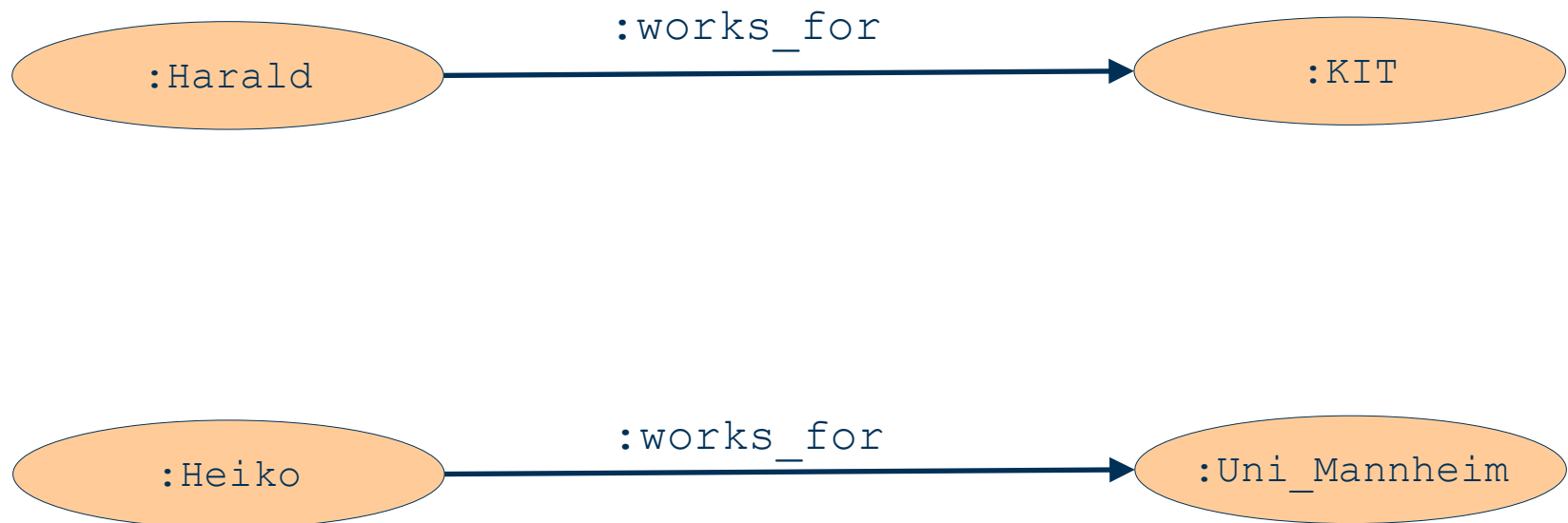
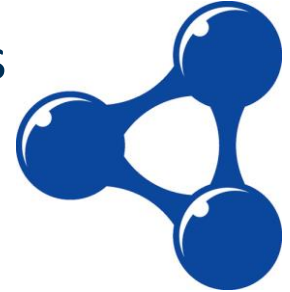
- „Resource Description Framework“
- A W3C standard since 2004
- Description of arbitrary things
- View 1: Sentences in the form <subject, predicate, object>
- „Heiko works for the University of Mannheim.“
- View 2: Directed graphs with labeled edges

Some literature:
<head, relation, tail>



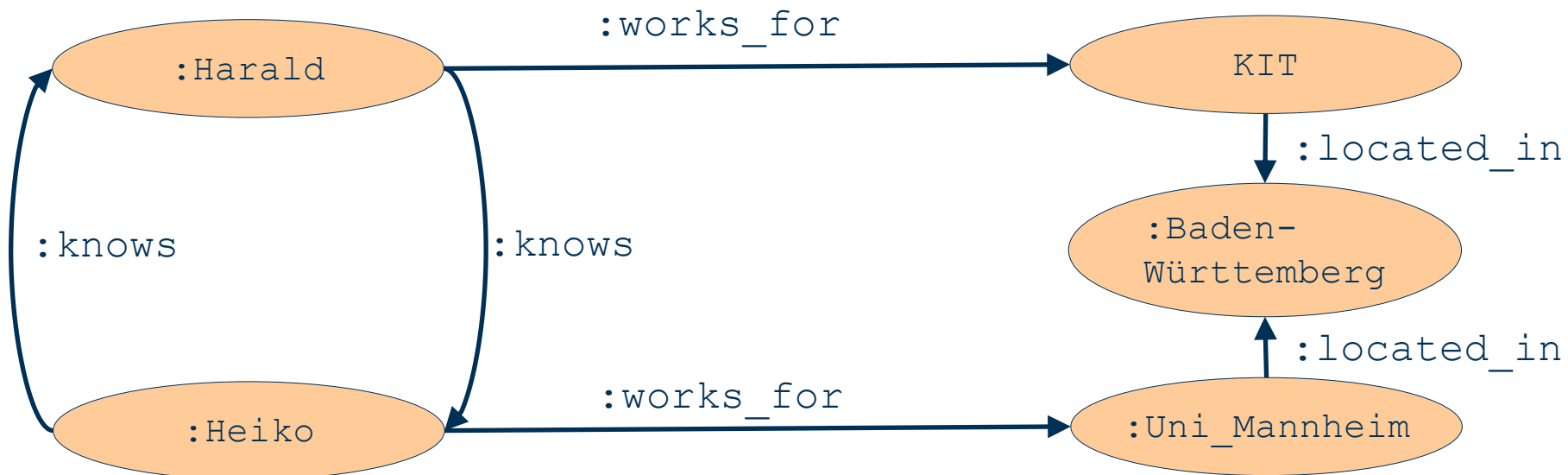
What is RDF?

- A knowledge graph consists of *multiple* sentences



What is RDF?

- We usually think of knowledge graphs as densely connected graphs
 - Objects of one statement become subjects of another

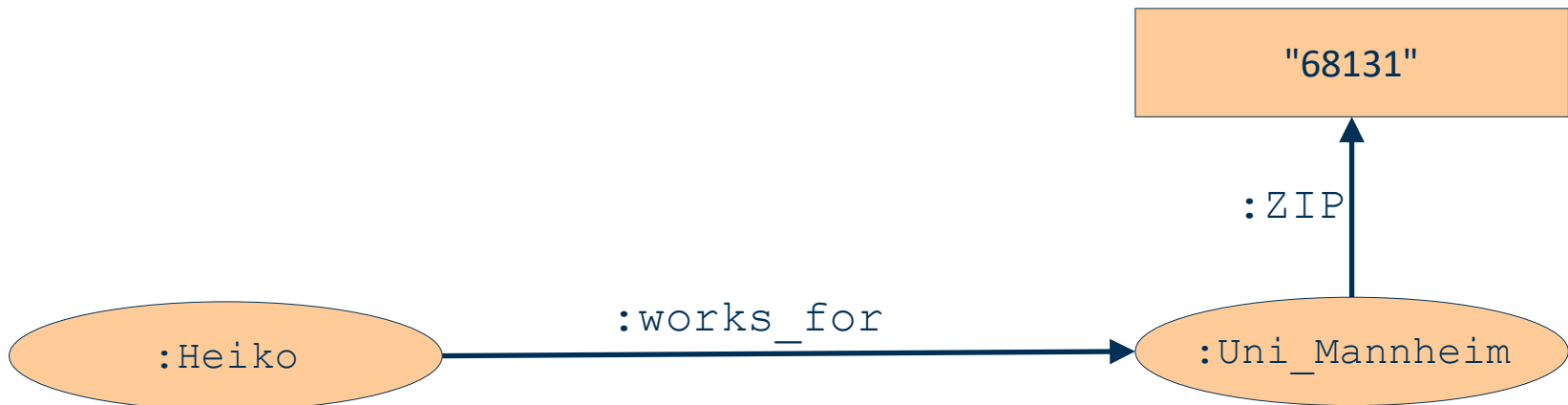


Basic Building Blocks of RDF

- Resources
 - denote things
 - are identified by a URI
 - can have one or multiple types
- Literals
 - are values like strings or integers
 - can only be objects, not subjects or predicates
(graph view: they can only have ingoing edges)
 - can have a datatype or a language tag (but not both)
- Properties (Predicates)
 - Link resources to other resources and to literals

Basic Building Blocks of RDF

- Resources
 - Heiko, Uni Mannheim
- Properties
 - works for, ZIP
- Literals
 - “68131”



Types

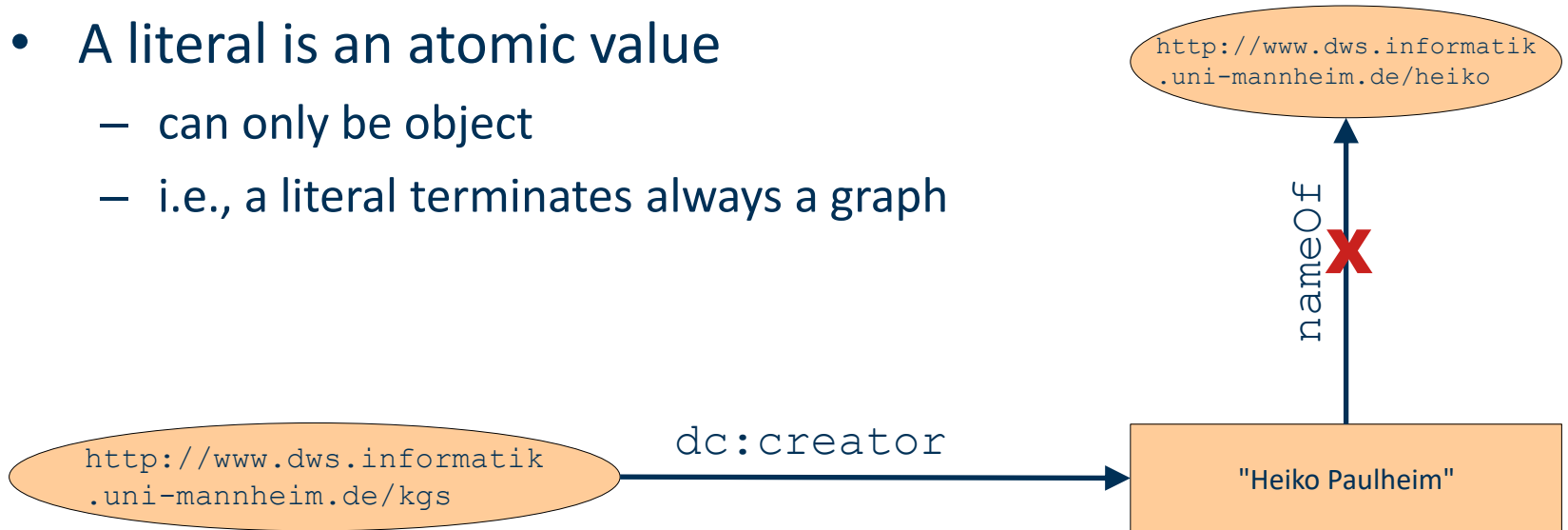
- All resources (not literals) can have a type
- Types can be arbitrarily defined
- The predefined predicate `rdf:type` (full URI below) defines the type of a resource
- Knowledge Graphs is a lecture



[<http://www.w3.org/1999/02/22-rdf-syntax-ns#type>](http://www.w3.org/1999/02/22-rdf-syntax-ns#type)

Resource vs. Literal

- A literal is an atomic value
 - can only be object
 - i.e., a literal terminates always a graph

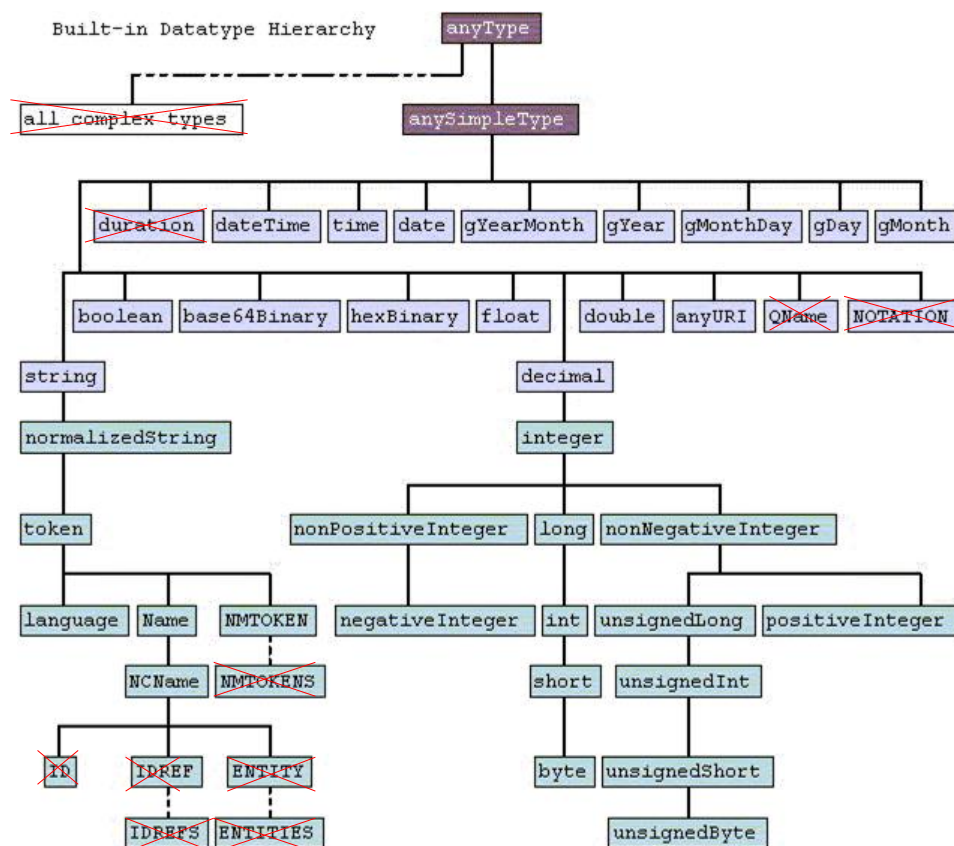


- A resource can be a subject itself



Datatypes for Literals

- (Almost) all XML Schema datatypes may be used
- Exception:
 - XML specific types
 - The underspecified type "duration"
 - sequence types



Language Tags for Literals

- Literals may be defined in different natural languages
 - "München"@de
 - "Munich"@en
- Those can be marked
- Note: Knowledge Graphs can be multilingual!
- Language codes according to ISO 963
 - ISO 963-1 (1963): two-digit codes, 136 languages
 - ISO 963-2 (1998): three-digit Codes, 464 languages
 - if both are defined, ISO 963-1 has to be used!
- http://www.loc.gov/standards/iso639-2/php/English_list.php



Datatypes in RDF

- Examples:

```
:Munich :hasName "München"@de .  
:Munich :hasName "Munich"@en .  
:Munich :hasPopulation "1356594"^^xsd:integer .  
:Munich :hasFoundingYear "1158-01-01"^^xsd:date .
```

- Note: there are no default datatypes (not even “string”!)

- These are three different literals:

- "München" @en
- "München"@de
- "München"^^xsd:string

Triple Notation

- A W3C standard (2004)
- Triples consist of a subject, predicate, and object
- An RDF document is an *unordered* set of triples



- Simple triple:

```
<http://www.dws.informatik.uni-mannheim.de/teaching/semantic-web>  
<http://purl.org/dc/elements/1.1/relation>  
<http://www.w3.org/2001/sw/> .
```

- Literal with language tag:

```
<http://www.dws.informatik.uni-mannheim.de/teaching/semantic-web>  
<http://purl.org/dc/elements/1.1/subject>  
"Semantic Web"@en .
```

- Type literal:

```
<http://www.dws.informatik.uni-mannheim.de/teaching/semantic-web>  
<http://www.uni-mannheim.de/mhb/creditpoints>  
"6"^^<http://www.w3.org/2001/XMLSchema#integer> .
```

Turtle Notation

- A simplified triple notation
- Central definition of namespaces (and a default namespace):

```
@prefix dc: <http://purl.org/dc/elements/1.1/>  
@prefix : <http://www.dws.informatik.uni-mannheim.de/teaching/>  
:semantic-web dc:subject "Semantic Web"@en .
```

- Triples sharing the same subject or subject+predicate:

```
:semantic-web dc:subject "Semantisches Web"@de ,  
                "Semantic Web"@en ;  
                dc:creator "Heiko Paulheim".
```

- Shorthand notation for `rdf:type`:

```
:semantic-web a :lecture .
```

Notation RDF/XML

- A W3C standard since 2004
- Encodes RDF in XML
- Suitable for machine processing (plenty of XML tools!)



- Defining resources

```
<rdf:Description rdf:about="http://dws.informatik.uni-mannheim.de/teaching/kgs">  
  <dc:creator>Heiko Paulheim</dc:creator>  
</rdf:Description>
```

- Defining typed resources

```
<rdf:Description rdf:about="http://dws.informatik.uni-mannheim.de/teaching/kgs">  
  <rdf:type rdf:resource="http://www.uni-mannheim.de/mhb/Lecture"/>  
</rdf:Description>
```

- Alternative representation

```
<mhb:Lecture rdf:about="http://dws.informatik.uni-mannheim.de/teaching/kgs"  
  xmlns:mhb="http://www.uni-mannheim.de/mhb/" />
```

Notation RDF/XML

- Relations between resources by nesting

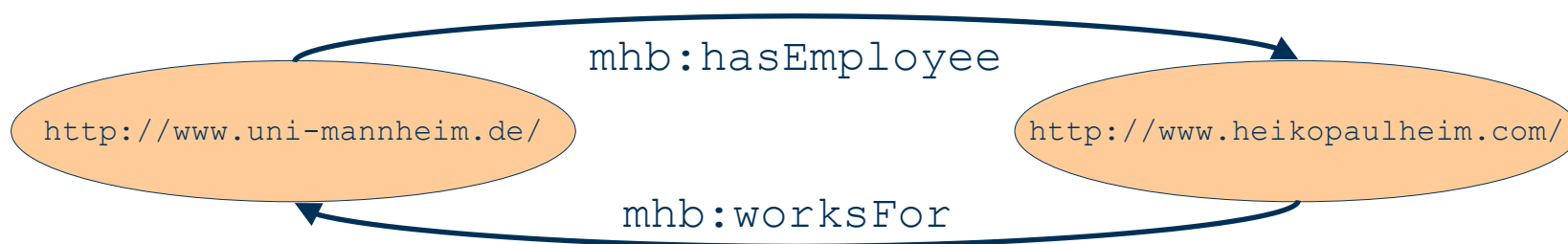
```
<mhb:Lecture rdf:about="http://dws.informatik.uni-mannheim.de/teaching/kgs">  
  <mhb:givenBy>  
    <mhb:Lecturer rdf:about="http://dws.informatik.uni-mannheim.de/heiko"/>  
  </mhb:givenBy>  
</mhb:Lecture>
```

- Relations between resources by explicit links

```
<mhb:Lecturer rdf:about="http://dws.informatik.uni-mannheim.de/heiko" />  
  
<mhb:Lecture rdf:about="http://dws.informatik.uni-mannheim.de/teaching/kgs">  
  <mhb:givenBy rdf:resource="http://dws.informatik.uni-mannheim.de/heiko"/>  
</mhb:Lecture>
```

Notation RDF/XML

- Recap: knowledge graphs may contain cycles
- XML is acyclic – we need to use explicit links



```
<mhb:University rdf:about="http://www.uni-mannheim.de">
  <mhb:hasEmployee>
    <mhb:UniversityMember rdf:about="http://www.heikopaulheim.com/">
      <mhb:worksFor rdf:resource="http://www.uni-mannheim.de"/>
    </mhb:UniversityMember>
  </mhb:hasEmployee>
</mhb:University>
```

JSON-LD Notation

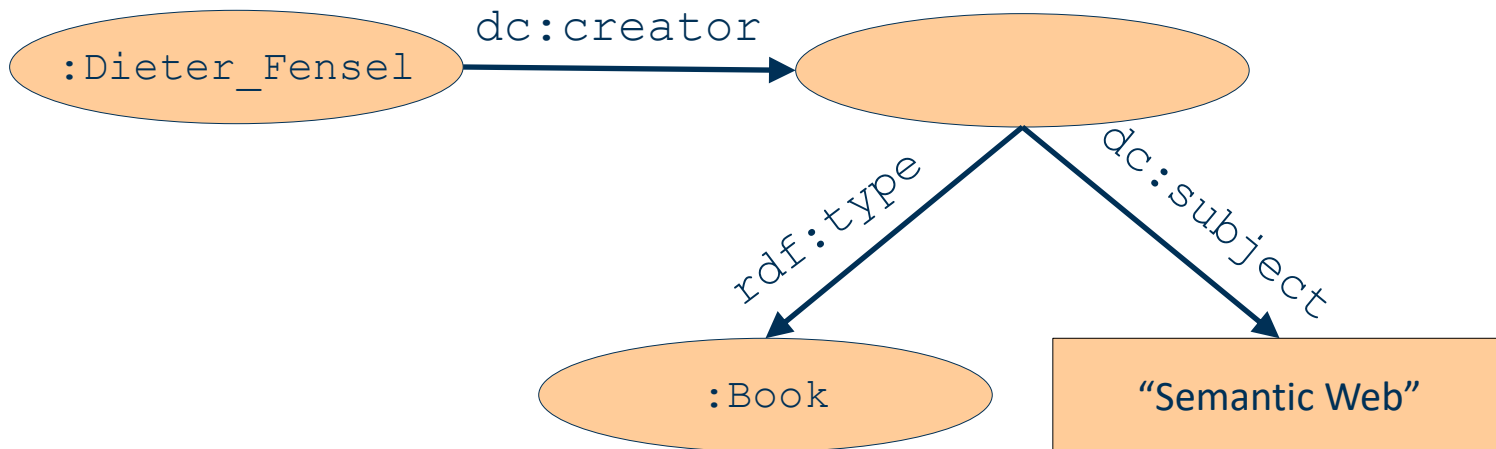
- JSON is popular in script programming
- JSON-LD: Standard for serializing RDF in JSON
 - JSON-LD 1.1 W3C standard (2020)



```
{
  "@id": "http://www.heikopaulheim.com/",
  "http://dws.informatik.uni-mannheim.de/name": "Heiko Paulheim",
  "http://dws.informatik.uni-mannheim.de/teaches":
  {
    "http://http://www.w3.org/1999/02/22-rdf-syntax-ns#type":
      "http://dws.informatik.uni-mannheim.de/Lecture",
    "http://dws.informatik.uni-mannheim.de/title":
      "Knowledge Graphs"
  }
}
```

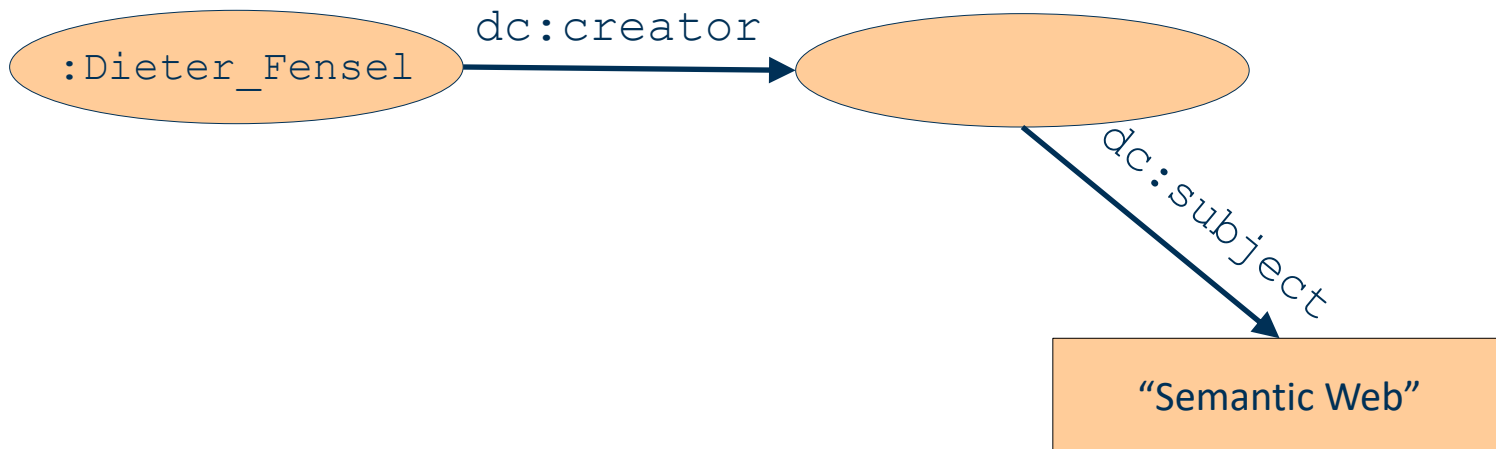
Blank Nodes

- Information that is not or cannot be specified
 - "Dieter Fensel has written a book about the Semantic Web"



Blank Nodes

- Information that is not or cannot be specified
 - "Dieter Fensel has written something about the Semantic Web."



Blank Nodes in Turtle

- Variant 1: explicitly named with an underscore

```
:Dieter_Fensel dc:creator _:x .  
_:x a :Book ;  
    dc:subject "Semantic Web" .
```

- Variant 2: unnamed with square brackets

```
:Dieter_Fensel dc:creator [  
    a :Book;  
    dc:subject "Semantic Web"  
].
```

- Notes
 - both are equivalent
 - changing blank node names does not change the semantics!

Application of Blank Nodes: n-ary Predicates

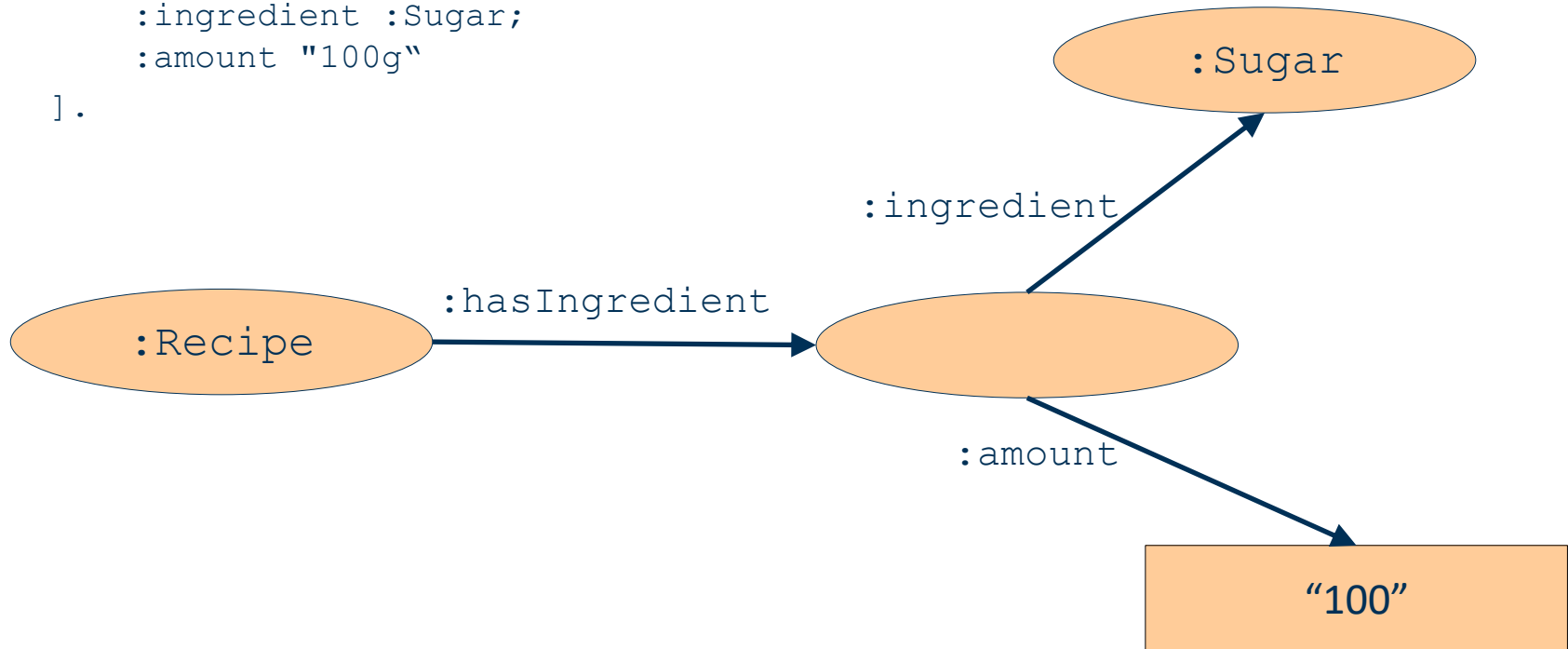
- RDF predicates always connect a subject and an object
 - i.e., in the sense of predicate logic, they are binary predicates

`:Heiko :works_for :UniMannheim .`
 $\Leftrightarrow$ `works_for(Heiko, UniMannheim) .`

- Sometimes, n-ary predicates are needed
 - `has_ingredient(Recipe, Sugar, 100g)`

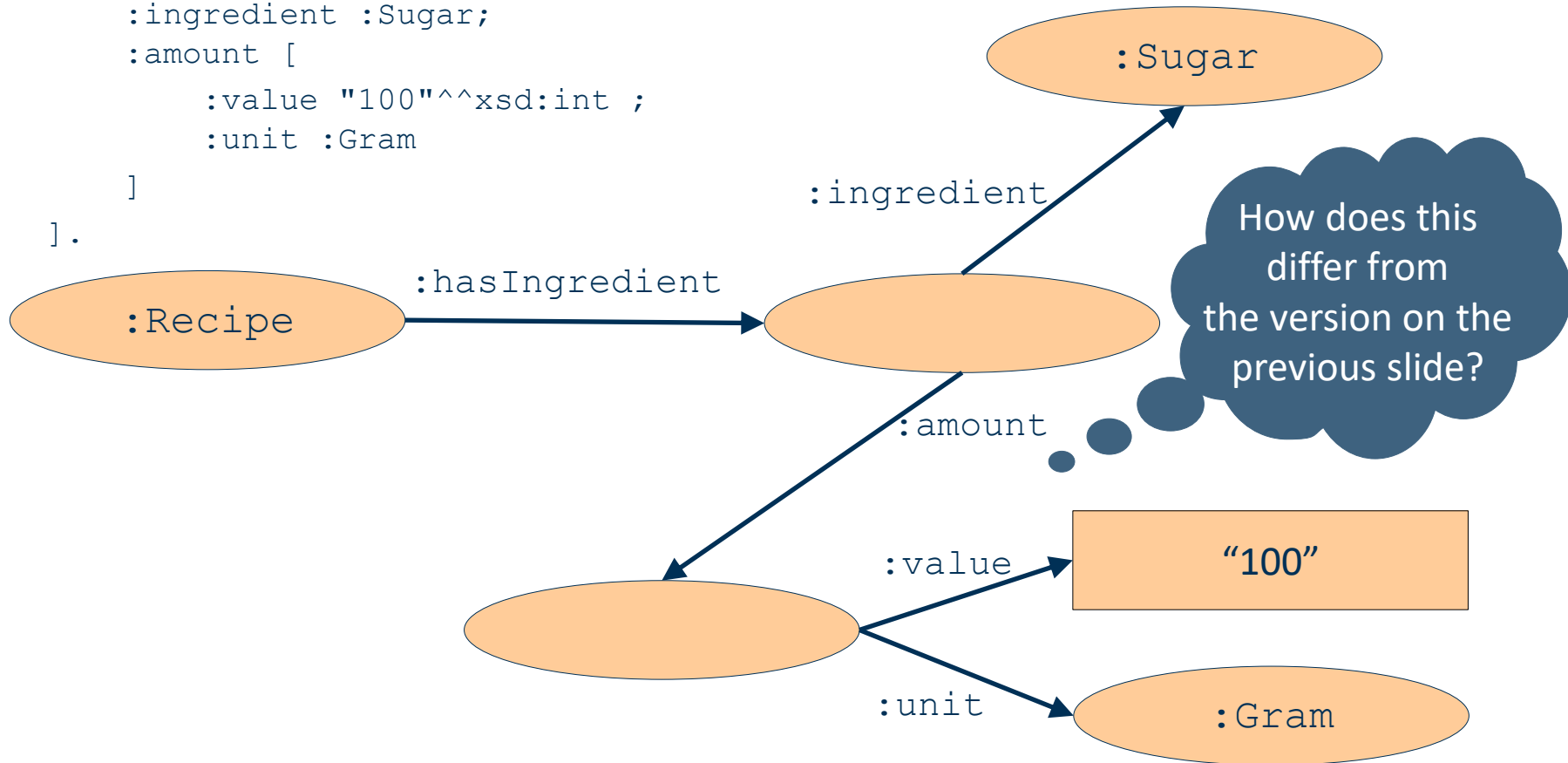
Application of Blank Nodes: n-ary Predicates

```
:Recipe :hasIngredient [  
  :ingredient :Sugar;  
  :amount "100g"  
].
```



Application of Blank Nodes: n-ary Predicates

```
:Recipe :hasIngredient [  
  :ingredient :Sugar;  
  :amount [  
    :value "100"^^xsd:int ;  
    :unit :Gram  
  ]  
].
```



Semantic Principles of RDF

- One thing can have multiple names

`:Munich :capitalOf :Bavaria .`

`:München :capitalOf :Bayern .`

- On the semantic web, there is not just one name for each thing
 - this is called the *Non-unique name assumption*
- This means: Just that two things have different names does not mean that they are different!

RDF: Intuition and Actual Semantics

- Let us consider the following example:

```
:Peter :fatherOf :Julia ,  
                :Mary .
```

- How many children does Peter have?
- Intuitively, we assume that Julia and Mary are two different persons
- However, this is not trivial for a machine
 - (and the assumption may even be wrong)

Semantic Principles of RDF

- Historical argumentation:
 - We (probably) do not know all the contents of the Semantic Web*
- Therefore, there may be more information on a resource than what we have
- This principle is called "Open World Assumption"

* this may not be true for enterprise knowledge graphs

RDF: Intuition and Actual Semantics

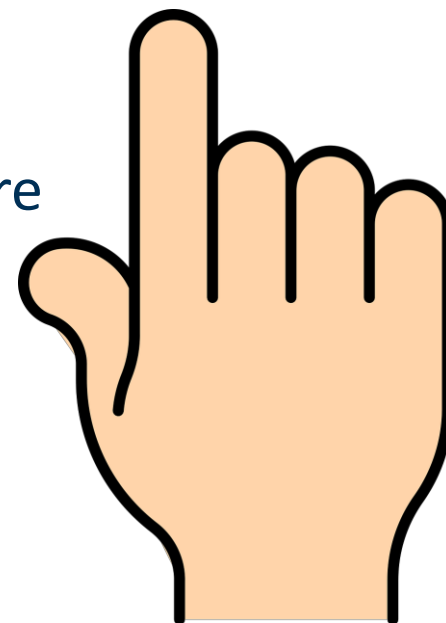
- Let us consider this example again:

```
:Peter :fatherOf :Julia ,  
                :Mary .
```

- How many children does Peter have?
- Intuition says: two children
- However, he could also have three or more
(or also just one, as we have learned just a minute ago)

RDF: Intuition and Actual Semantics

- Both
 - Non-unique Name Assumption and
 - Open World Assumption
- will re-occur quite a few times in this lecture
- Hint: consider those two whenever something seems weird when interpreting RDF data



RDF and HTML

- The Semantic Web uses RDF
- The “classic” Web uses HTML
- Does that mean that each information has to be encoded twice?
 - once for humans, once for machines

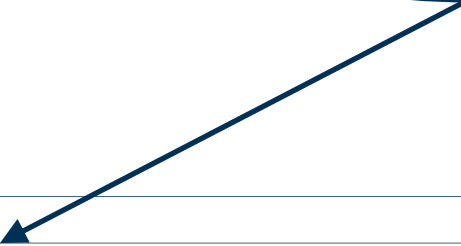
```
<html>
...
<b>Dr. Mark Smith</b>
<i>Physician</i>
Main St. 14
Smalltown
Mon-Fri 9-11 am
Wed 3-6 pm
...
</html>
```

```
:p a :Physician .
:p :hasDegree "Dr." .
:p :hasName "Mark Smith" .
:p :hasAddress :a .
:a :street "Main Street" .
:a :number "14"^^xsd:int .
:a :city "Smalltown" .
:p :hasOpeningHours [
  a rdf:Bag ;
  [ :day :Monday;
    :from "9"^^xsd:int;
    :to "11"^^xsd:int;
  ]
  ...
]
```

Using RDF and HTML Together – Variant 1

- Explicit reference to a RDF version
 - an agent stumbling on the HTML page can download the RDF data file

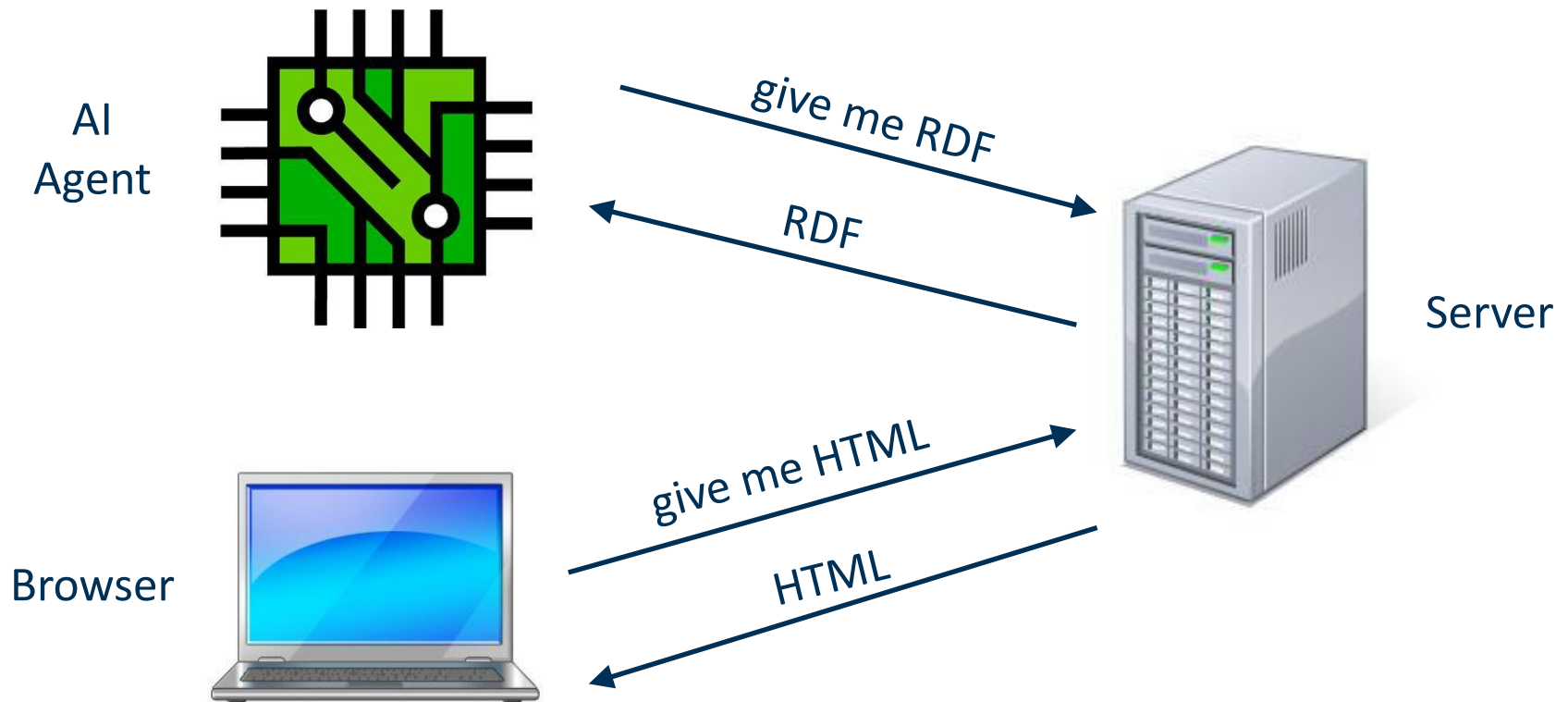
```
<html>
  <head>
    <link rel="meta" type="application/rdf+xml" title="DC" href="dc.rdf"/>
  </head>
  <body>
    ...
  </body>
</html>
```



```
<?xml version="1.0"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:dc="http://purl.org/dc/elements/1.1/">
  <rdf:Description rdf:about="http://www.dws.informatik.uni-mannheim.de/mhb/sw">
    <dc:publisher>University of Mannheim</dc:publisher>
    <dc:subject>Semantic Web</dc:subject>
    <dc:creator>Heiko Paulheim</dc:creator>
    <dc:relation rdf:resource="http://www.w3.org/2001/sw/" />
  </rdf:Description>
</rdf:RDF>
```

Using RDF and HTML Together – Variant 2

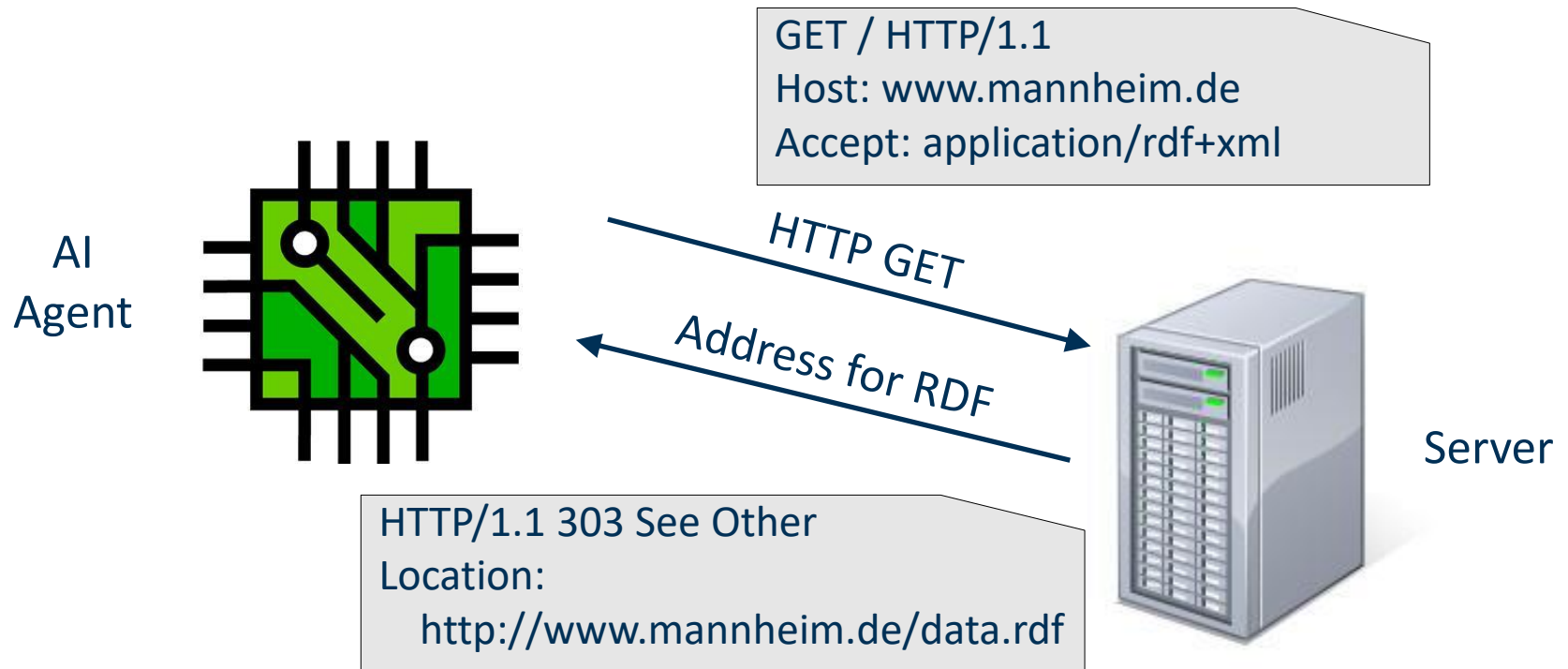
- Content Negotiation



Content Negotiation in Detail



Content Negotiation in Detail



Content Negotiation: MIME Types

- MIME: Multipurpose Internet Mail Extensions
- Original purpose: classifying e-mail attachments
 - Text, PDF, ..
- First version: 1996
- Administrated by IANA
- Important MIME types for Semantic Web programming
 - application/rdf+xml
 - text/turtle
 - text/n3
 - application/json
 - application/sparql-query
 - application/sparql-results+xml

Using RDF and HTML Together

- Link to RDF Document
 - Can be done with a simple HTML editor
 - No special server configuration needed
- Content Negotiation
 - Requires particular server setup
 - *One* URI can be used for different representations
- Both cases require
 - two different representations
 - “double bookkeeping”

→ Potential source of inconsistencies!

RDF in Attributes (RDFa)

- Idea of RDFa
 - Why not encode HTML and RDF in *one* document
 - The essential information only has to be encoded once
- RDFa combines XHTML with RDF
 - W3C Standard since 2008



```
<html>
...
<body about="http://www.marcsmith.com/MarcSmith">
  <b><span property="doc:name">Dr. Mark Smith</span></b>
  <i><span property="doc:profession">Physician</span></i>
  <span rel="doc:address" href="http://www.marcsmith.com/Address">
    <span property="doc:street">Main Street</span>
    <span property="doc:number">14</span>
    <span property="doc:city">Smalltown</span>
  </span>
  ...
</body>
</html>
```

RDFa Language Constructs

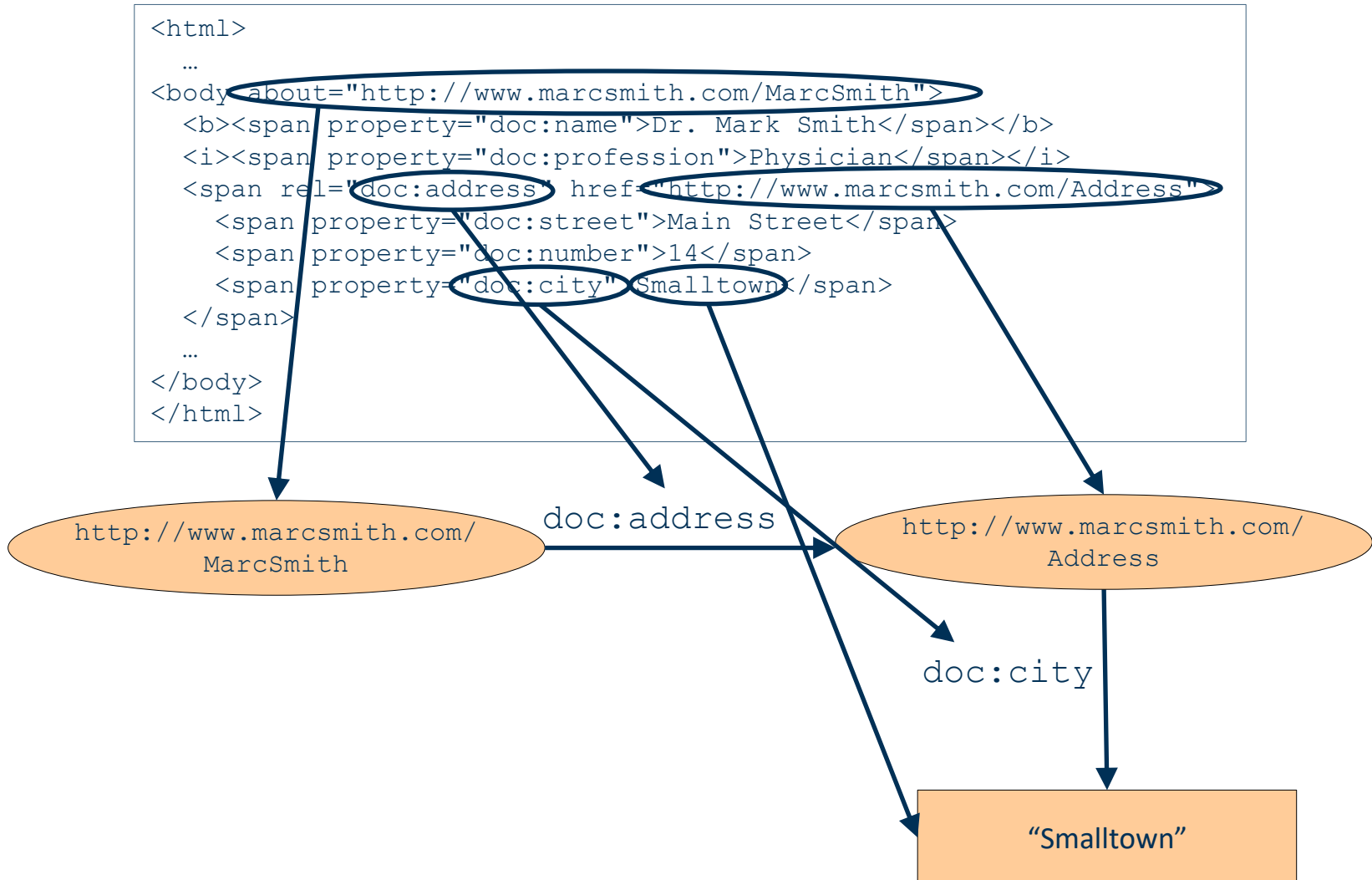
- `about="http://foo.bar/aSubject"`
 - Defines the subject of a page or section
- `property="http://foo.bar/aProperty"`
 - Defines a relation
 - Contents of the tag are interpreted as a literal
- `rel="http://foo.bar/aRelation"`
 - Defines a relation to another resource
- `href="http://foo.bar/aResource"`
 - Defines a relation's object
 - can be the subject of a resource again
- `typeof="http://foo.bar/aType"`
 - defines a resource's type

RDF in Attributes (RDFa)

```
<html>
...
<body about="http://www.marcsmith.com/MarcSmith">
  <b><span property="doc:name">Dr. Mark Smith</span></b>
  <i><span property="doc:profession">Physician</span></i>
  <span rel="doc:address" href="http://www.marcsmith.com/Address">
    <span property="doc:street">Main Street</span>
    <span property="doc:number">14</span>
    <span property="doc:city">Smalltown</span>
  </span>
  ...
</body>
</html>
```



RDF in Attributes (RDFa)



Alternative to RDFa: Microdata

- Adding structured information to web pages
 - By marking up contents
 - Arbitrary vocabularies are possible
 - Introduced with HTML5
- Similar to RDFa
 - W3C Standard since 2008, maintained by WHATWG since 2019



```
<div itemscope itemtype="http://schema.org/PostalAddress">  
  <span itemprop="name">Data and Web Science Group</span>  
  <span itemprop="addressLocality">Mannheim</span>,&br/>  <span itemprop="postalCode">68131</span>  
  <span itemprop="addressCountry">Germany</span>  
</div>
```

Alternative to RDFa: Microdata



- Markup can be extracted to RDF
 - See W3C Interest Group Note: Microdata to RDF [1]

```
<div itemscope itemtype="http://schema.org/PostalAddress">  
  <span itemprop="name">Data and Web Science Group</span>  
  <span itemprop="addressLocality">Mannheim</span>,&br/>  <span itemprop="postalCode">68131</span>  
  <span itemprop="addressCountry">Germany</span>  
</div>
```

```
_:1 a <http://schema.org/PostalAddress> .  
_:1 <http://schema.org/name> "Data and Web Science Group".  
_:1 <http://schema.org/addressLocality> "Mannheim" .  
_:1 <http://schema.org/postalCode> "68131" .  
_:1 <http://schema.org/adressCountry> "Germany" .
```

Alternative to RDFa: Microdata

- Commonalities
 - Arbitrary classes/predicates are possible
 - Although Microdata is mainly used with schema.org
- Differences
 - Microdata is slightly less expressive
 - No URIs, only blank nodes
 - No cycles in the resulting RDF graph
 - No reification (see later)

RDFa, MicroFormats, and Microdata

- In a specific `<script>` tag
- Note: unlike RDFa/Microdata, this is content duplication!

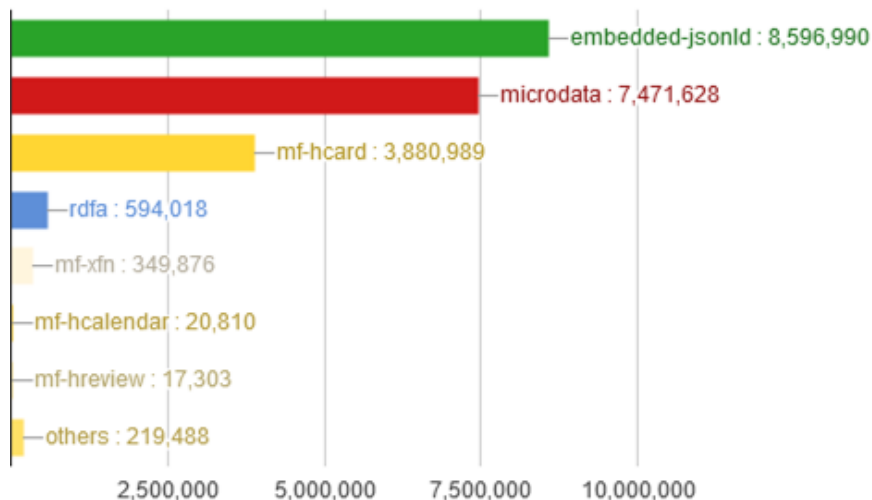
```
<script>
  "@context": "https://schema.org",
  "@type": "Organization",
  "address": {
    "@type": "PostalAddress",
    "name": "Data and Web Science Group",
```

```
_:0 a <http://schema.org/Organization> .
_:0 <http://schema.org/address> _:1 .
_:1 a <http://schema.org/PostalAddress> .
_:1 <http://schema.org/name> "Data and Web Science Group" .
_:1 <http://schema.org/addressLocality> "Mannheim" .
_:1 <http://schema.org/postalCode> "68131" .
_:1 <http://schema.org/adressCountry> "Germany" .
```

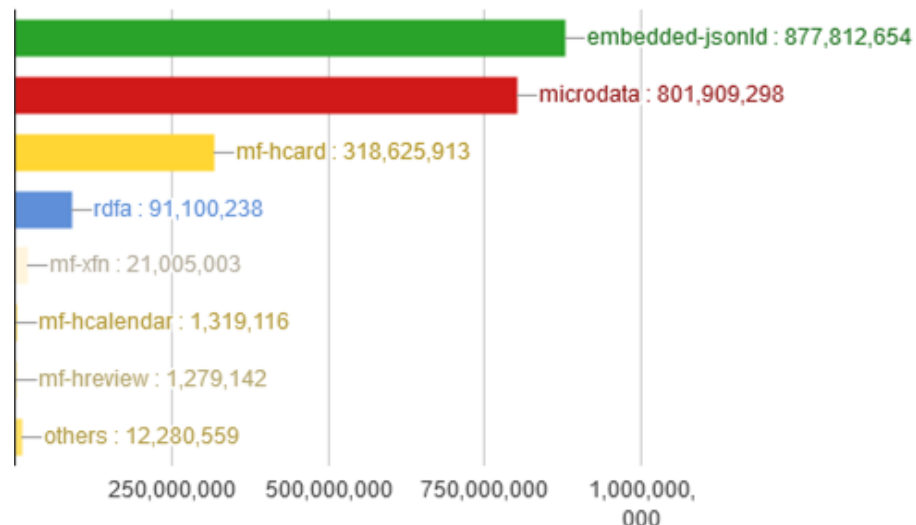
RDFa, MicroFormats, and Microdata

- MicroFormats: fixed vocabularies for persons, addresses, etc.
- WebDataCommons: Large-Scale Extraction of RDFa, MicroFormats, Microdata, JSON-LD from the Web

Domains with Triples



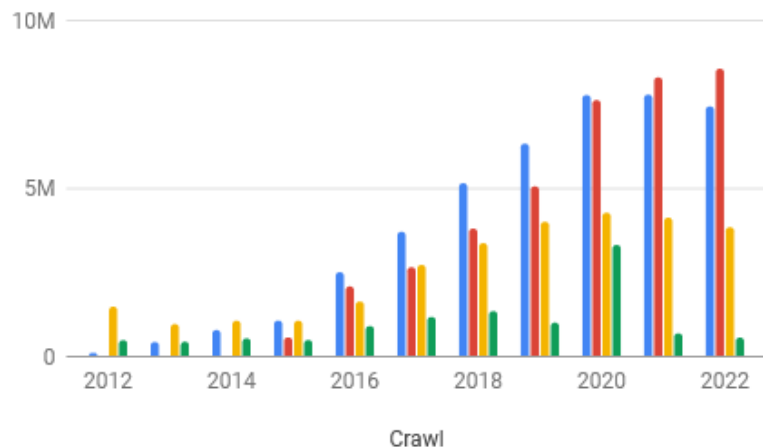
URLs with Triples



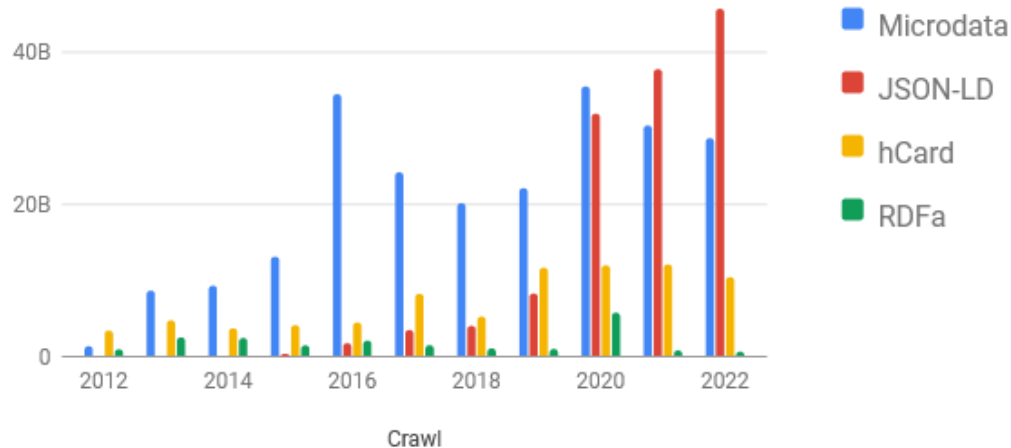
RDFa, MicroFormats, and Microdata

- MicroFormats: fixed vocabularies for persons, addresses, etc.
- WebDataCommons: Large-Scale Extraction of RDFa, MicroFormats, Microdata, JSON-LD from the Web

Number of PLDs Deploying the four Major Markup Formats

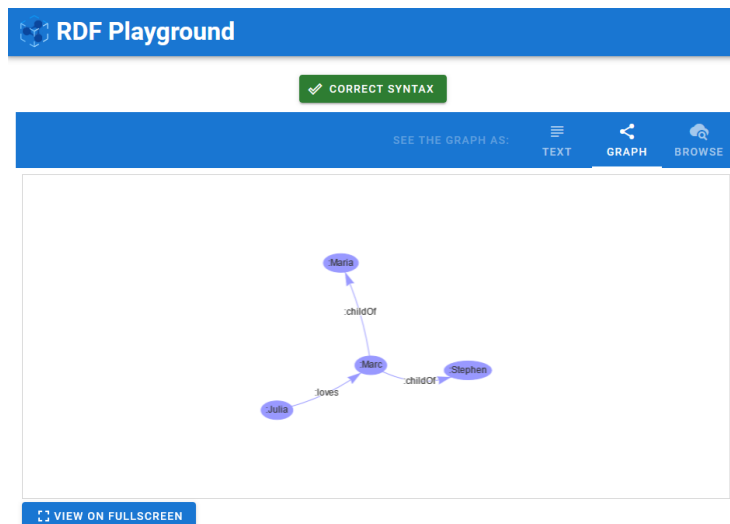


Number of Triples marked up by the four Major Markup Formats



RDF Tools

- Storage: relational databases, graph databases, ...
- Validation: validating parsers checking consistency
- Visualization: mostly graph based visualization
- Reasoning: inference over graphs (see next lectures)
- Programming: APIs (see next lectures)



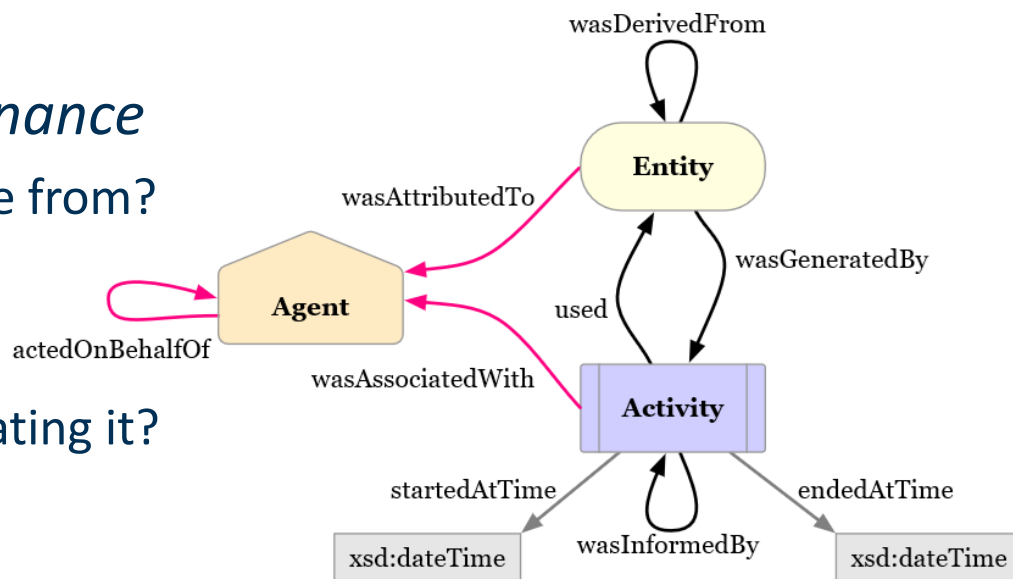
<http://rdfplayground.dcc.uchile.cl/>

Metadata for RDF

- Recap: Dublin Core was designed as Metadata for the Web
- Knowledge graphs may have metadata as well

- Most prominently: *provenance*

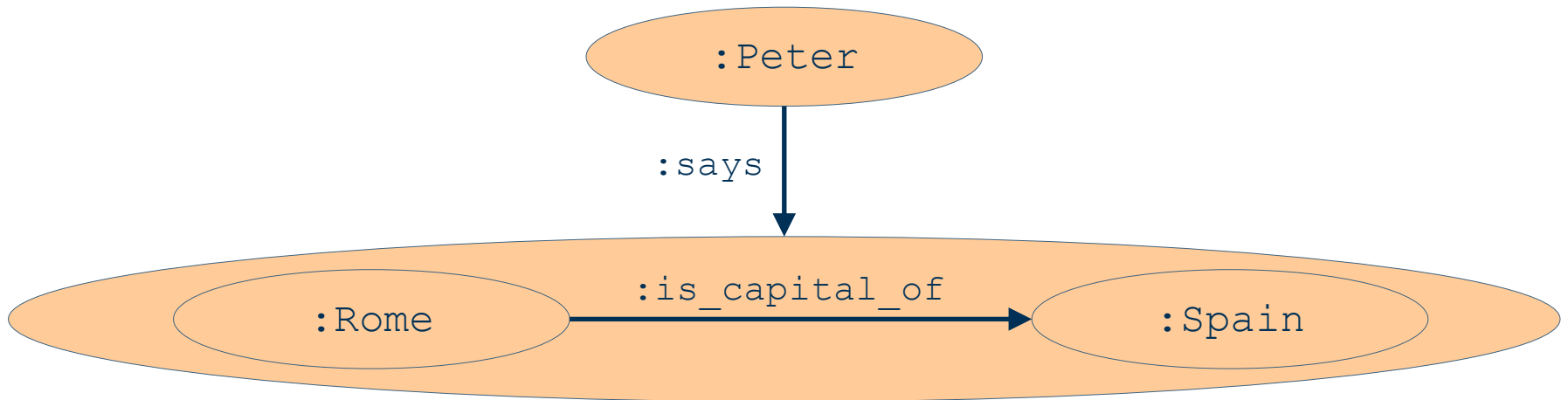
- Where does the data come from?
- Who created it?
- When was it created?
- What was the process creating it?
- ...



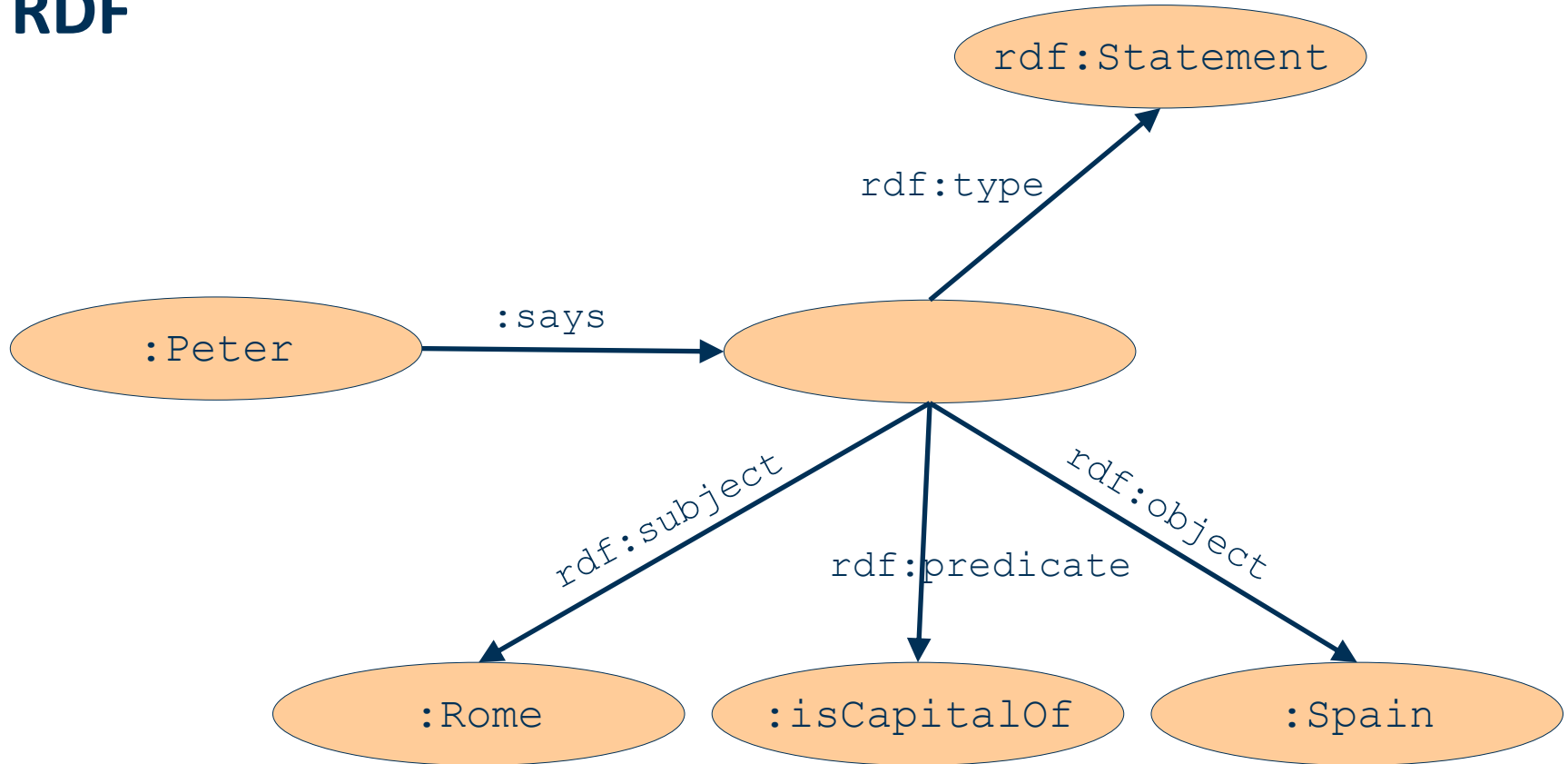
Reification

- Latin *res* ("Thing"), *facere* ("make")
 - An Explication
 - making a statement, an opinion etc. the subject of a statement
- In RDF: Statements about statements
- "Peter says that Rome is the capital of Spain."
- Implementation:
 - RDF Statements are considered resources themselves
 - Can be subject or object of other statements
- Reification can have multiple levels
 - "Peter says that Wikipedia states that Rome is the capital of Spain."

Reification in RDF



Implementing Reification as Standard RDF



Encoding Reification in Turtle

- Variant 1: Named Statement (with URI)

```
:triple1 rdf:type rdf:Statement ;  
          rdf:subject :Rome ;  
          rdf:predicate :isCapitalOf ;  
          rdf:object :Spain .  
:Peter :says :triple1 .
```

- Variant 2: Unnamed Statement (Blank Node)

```
:Peter :says [  
  a rdf:Statement ;  
  rdf:subject :Rome ;  
  rdf:predicate :isCapitalOf ;  
  rdf:object :Spain .  
] .
```

Wrap Up

- RDF is a language for describing arbitrary things
 - interpretation: set of statements or directed graph
 - Notations: RDF/XML, Turtle
- Special language constructs
 - Blank nodes
 - Reification and its variants
- Semantics
 - Non-unique name assumption
 - Open world assumption
- Embedding in HTML is possible
- Large set of tools is available

A Critical Look in the Rear View Mirror

- Is RDF more powerful than XML?
- XML is a markup language for information
- In XML, arbitrary elements and attributes can be defined
- XML tag names are meaningless for a computer
- RDF is a markup language for information
- In RDF, arbitrary classes and predicates can be defined
- RDF class and predicate names are meaningless for a computer

A Critical Look in the Rear View Mirror

- So, why did we spend an entire lecture on RDF?



A Critical Look in the Rear View Mirror

HOW STANDARDS PROLIFERATE:
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC.)



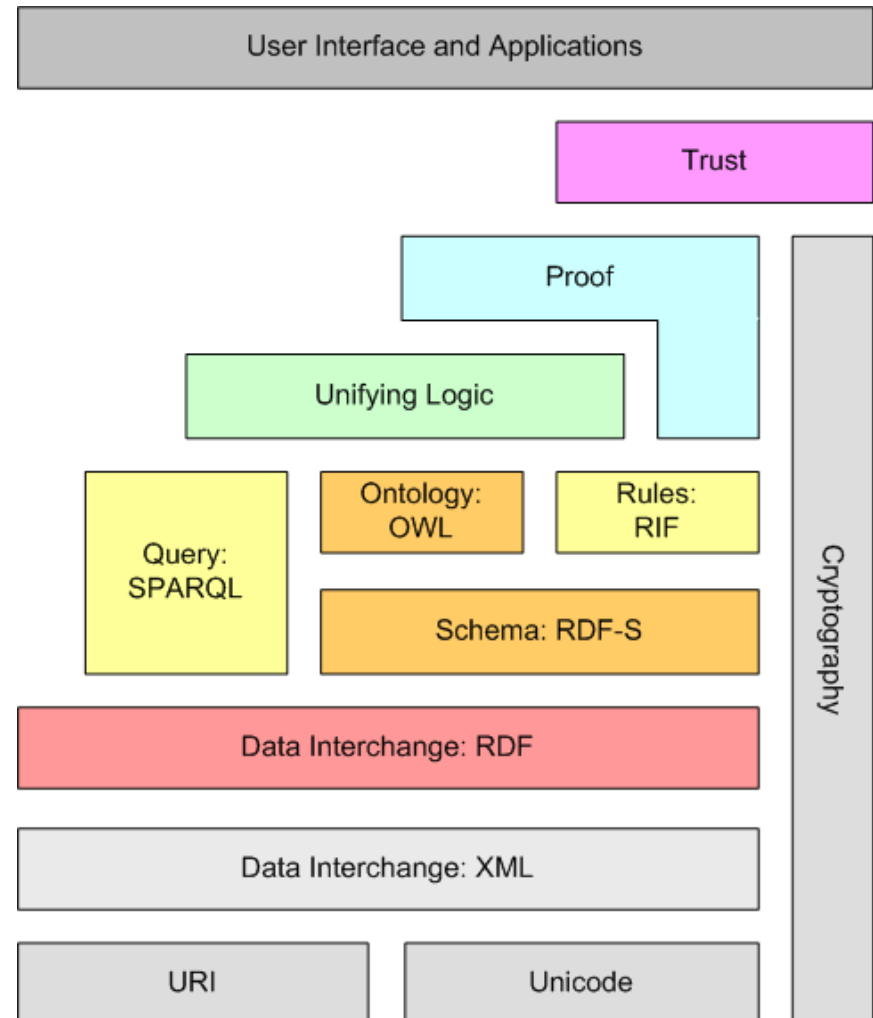
The Semantic Web Stack



here be dragons...

Knowledge Graph Technologies
(This lecture)

Technical
Foundations



Questions?

